

# Complex Analysis: Contours, Fourier Transforms, and Schwarz–Christoffel

Problems with background and complete solutions

## 1 Problem 1: Evaluate $\int_{-1}^1 \frac{\sqrt{1-x^2}}{x^2+1} dx$ by contour integration

*Hint.* Consider  $F(z) = \sqrt{z^2-1}/(z^2+1)$  with a branch cut on  $[-1, 1]$ , and integrate around a large circle.

### Background

Fix the branch of  $G(z) = \sqrt{z^2-1}$  with cut on  $[-1, 1]$  and  $G(x) > 0$  for  $x > 1$ . For  $x \in (-1, 1)$ ,

$$G(x+i0) = +i\sqrt{1-x^2}, \quad G(x-i0) = -i\sqrt{1-x^2}.$$

Hence the sum of the upper and lower lip integrals across the cut equals

$$2i \int_{-1}^1 \frac{\sqrt{1-x^2}}{x^2+1} dx = 2i I.$$

### Solution

Define

$$F(z) = \frac{\sqrt{z^2-1}}{z^2+1}, \quad z \in \mathbb{C} \setminus [-1, 1].$$

Poles:  $z = \pm i$  (simple). Large-circle behavior:  $F(z) = \frac{1}{z} + O(z^{-3})$ ; thus

$$\int_{|z|=R} F(z) dz \longrightarrow 2\pi i \quad (R \rightarrow \infty).$$

Deforming the big circle to a contour hugging the cut gives

$$\underbrace{\int_{|z|=R} F(z) dz}_{\rightarrow 2\pi i} = \underbrace{\int_{\text{upper lip}} F dz + \int_{\text{lower lip}} F dz}_{= 2i I} + (\text{vanishing end caps}).$$

Residues at the simple poles:

$$\begin{aligned} \operatorname{Res}_{z=i} F &= \lim_{z \rightarrow i} \frac{(z-i)\sqrt{z^2-1}}{(z-i)(z+i)} = \frac{\sqrt{i^2-1}}{2i} = \frac{\sqrt{-2}}{2i} = \frac{\sqrt{2}}{2}, \\ \operatorname{Res}_{z=-i} F &= \lim_{z \rightarrow -i} \frac{(z+i)\sqrt{z^2-1}}{(z-i)(z+i)} = \frac{\sqrt{(-i)^2-1}}{-2i} = \frac{\sqrt{-2}}{-2i} = \frac{\sqrt{2}}{2}. \end{aligned}$$

Hence, by the residue theorem,

$$\int_{|z|=R} F(z) dz \longrightarrow 2\pi i \left( \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} \right) = 2\pi i \sqrt{2}.$$

Equating the two expressions for the big-circle integral,

$$2i I = 2\pi i (\sqrt{2} - 1) \quad \Rightarrow \quad \boxed{I = \pi(\sqrt{2} - 1)}.$$

**Check (real calculus)**

With  $x = \sin \theta$ ,

$$I = 2 \int_0^{\pi/2} \frac{\cos^2 \theta}{1 + \sin^2 \theta} d\theta = 2 \int_0^\infty \frac{dt}{(1+t^2)(1+2t^2)} dt = \pi(\sqrt{2} - 1).$$


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**2 Problem 2: Fourier transforms and inverse verification**

We use the convention  $\widehat{f}(\xi) = \int_{-\infty}^\infty f(x) e^{-i\xi x} dx$  and  $f(x) = \frac{1}{2\pi} \int_{-\infty}^\infty \widehat{f}(\xi) e^{i\xi x} d\xi$ .

**(a)  $f(x) = e^{-|x|}$  — Transform and step-by-step inverse**

*Transform:*

$$\widehat{f}(\xi) = \int_0^\infty e^{-(1+i\xi)x} dx + \int_{-\infty}^0 e^{(1-i\xi)x} dx = \frac{1}{1+i\xi} + \frac{1}{1-i\xi} = \boxed{\frac{2}{1+\xi^2}}.$$

*Inverse by residues (two cases):*

$$f(x) = \frac{1}{2\pi} \int_{\mathbb{R}} \frac{2e^{i\xi x}}{1+\xi^2} d\xi = \frac{1}{2\pi} \int_{\mathbb{R}} \frac{2e^{i\xi x}}{(\xi-i)(\xi+i)} d\xi.$$

**Case  $x > 0$ :**  $f(x) = \frac{1}{2\pi} \oint_{\Gamma^+} \frac{2e^{i\xi x}}{(\xi-i)(\xi+i)} d\xi$  (close upward; Jordan's lemma)

$$= \frac{1}{2\pi} (2\pi i) \operatorname{Res}_{\xi=i} \left( \frac{2e^{i\xi x}}{(\xi-i)(\xi+i)} \right) = \frac{i 2e^{ix}}{2i(i+i)} = e^{-x}.$$

**Case  $x < 0$ :**  $f(x) = \frac{1}{2\pi} \oint_{\Gamma^-} \frac{2e^{i\xi x}}{(\xi-i)(\xi+i)} d\xi$  (close downward)

$$= \frac{1}{2\pi} (2\pi i) \operatorname{Res}_{\xi=-i} \left( \frac{2e^{i\xi x}}{(\xi-i)(\xi+i)} \right) = e^x.$$

Therefore  $\boxed{f(x) = e^{-|x|}}$ .

**(b)  $f(x) = e^{-a^2 x^2}$  ( $a > 0$ ) — Transform and step-by-step inverse**

*Transform:*

$$\widehat{f}(\xi) = \int_{\mathbb{R}} e^{-a^2 x^2 - i\xi x} dx = e^{-\xi^2/(4a^2)} \int_{\mathbb{R}} e^{-a^2 \left(x + i\frac{\xi}{2a^2}\right)^2} dx$$

$$\Rightarrow \boxed{\widehat{f}(\xi) = \frac{\sqrt{\pi}}{a} e^{-\xi^2/(4a^2)}}.$$

(Justify by integrating over a large rectangle whose vertical sides are the real line and the line shifted up by  $i\xi/(2a^2)$ ; the horizontal sides vanish by Gaussian decay.)

Inverse by a rectangular shift (explicit steps):

$$\begin{aligned}
 f(x) &= \frac{1}{2\pi} \int_{\mathbb{R}} \frac{\sqrt{\pi}}{a} e^{-\xi^2/(4a^2)} e^{i\xi x} d\xi \\
 &= \frac{1}{2\pi} \cdot \frac{\sqrt{\pi}}{a} \int_{\mathbb{R}} \exp\left(-\frac{1}{4a^2}(\xi - 2ia^2x)^2\right) \exp(-a^2x^2) d\xi \quad (\text{complete the square}) \\
 &= \frac{1}{2\pi} \cdot \frac{\sqrt{\pi}}{a} e^{-a^2x^2} \int_{\mathbb{R}+2ia^2x} \exp\left(-\frac{u^2}{4a^2}\right) du \quad (\text{shift contour vertically}) \\
 &= \frac{1}{2\pi} \cdot \frac{\sqrt{\pi}}{a} e^{-a^2x^2} \int_{\mathbb{R}} \exp\left(-\frac{u^2}{4a^2}\right) du \quad (\text{no poles; horizontal sides vanish}) \\
 &= \frac{1}{2\pi} \cdot \frac{\sqrt{\pi}}{a} e^{-a^2x^2} \cdot 2a\sqrt{\pi} = e^{-a^2x^2}.
 \end{aligned}$$


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### 3 Problem 3: Schwarz–Christoffel map from the half-plane to a rectangle

#### Statement

Write down, as an integral, the Schwarz–Christoffel map from the upper half-plane  $\mathbb{H}$  to a rectangle, with vertices being the images of the points  $z = \pm 1$  and  $z = \pm a$  (with  $a > 1$  real). Explain why  $a$  cannot be chosen arbitrarily, but is determined by the rectangle's aspect ratio.

#### Background

For prevertices  $x_k \in \mathbb{R}$  mapping to interior angles  $\alpha_k\pi$  in the polygon,

$$f'(z) = C \prod_k (z - x_k)^{\alpha_k - 1} \quad (\text{up to an affine change of variables}).$$

A rectangle has  $\alpha_k = \frac{1}{2}$  at each vertex, so every factor's exponent is  $-\frac{1}{2}$ .

#### Solution

Place prevertices at  $-a, -1, 1, a$  (in order) with  $a > 1$ . Then

$$f'(z) = C (z+a)^{-1/2} (z+1)^{-1/2} (z-1)^{-1/2} (z-a)^{-1/2} = \frac{C}{\sqrt{(z^2-1)(z^2-a^2)}}.$$

Integrating,

$$f(z) = A + C \int^z \frac{dt}{\sqrt{(t^2-1)(t^2-a^2)}}$$

maps  $\mathbb{H}$  conformally to a rectangle (up to translation and scaling).

**Aspect ratio determines  $a$** 

Side lengths are proportional to real integrals between consecutive prevertices:

$$W(a) = \int_{-1}^1 \frac{dt}{\sqrt{(1-t^2)(a^2-t^2)}}, \quad H(a) = \int_1^a \frac{dt}{\sqrt{(t^2-1)(a^2-t^2)}}.$$

Let  $k = 1/a \in (0, 1)$ . Standard substitutions give

$$W(a) = \frac{2}{a} K(k), \quad H(a) = \frac{2}{a} K'(k),$$

where  $K$  is the complete elliptic integral of the first kind and  $K'(k) = K(\sqrt{1-k^2})$ . Thus

$$\frac{H(a)}{W(a)} = \frac{K'(k)}{K(k)} \quad \text{with } k = \frac{1}{a}.$$

Given a rectangle, this ratio fixes  $k$  (hence  $a$ ) uniquely; therefore  $a$  is not free.

**4(a). Statement and branch points**

Suppose  $\zeta^2 = z^2 + 1$  and

$$z - i = r_1 e^{i\theta_1}, \quad z + i = r_2 e^{i\theta_2} \quad (r_1, r_2 > 0).$$

Then

$$\boxed{\zeta = \pm (r_1 r_2)^{1/2} e^{i(\theta_1 + \theta_2)/2}}.$$

*Reason.*  $z^2 + 1 = (z - i)(z + i) = r_1 r_2 e^{i(\theta_1 + \theta_2)}$ . Any square root has the two signs above. The branch points of the multifunction  $(z^2 + 1)^{1/2}$  are where either factor vanishes, i.e.  $z = \pm i$ .

**4(b). A branch with a finite vertical cut**

Define the branch by

$$f(z) = (r_1 r_2)^{1/2} e^{i(\theta_1 + \theta_2)/2}, \quad -\frac{\pi}{2} < \theta_1, \theta_2 \leq \frac{3\pi}{2}.$$

**Values across the imaginary axis**

Let  $z = x + iy$  and set  $\phi(y) = \frac{1}{2}(\theta_1 + \theta_2)$ . Note

$$r_1 = |y - 1|, \quad r_2 = |y + 1|, \quad \arg(it) = \begin{cases} \pi/2, & t > 0, \\ -\pi/2, & t < 0. \end{cases}$$

(i)  $|y| < 1$ .  $y - 1 < 0$ ,  $y + 1 > 0$ .

$$\begin{aligned} x \rightarrow 0^+ : \theta_1 \rightarrow (-\pi/2)^+, \theta_2 \rightarrow (\pi/2)^- &\Rightarrow \phi(y) \rightarrow 0, \\ x \rightarrow 0^- : \theta_1 \rightarrow (3\pi/2)^-, \theta_2 \rightarrow (\pi/2)^+ &\Rightarrow \phi(y) \rightarrow \pi. \end{aligned}$$

Hence

$$f(0^+ + iy) = \sqrt{1 - y^2}, \quad f(0^- + iy) = -\sqrt{1 - y^2}.$$

(ii)  $y > 1$ .  $\theta_1 = \theta_2 = \pi/2$  on both sides, so  $\phi(y) = \pi/2$  and

$$f(\pm 0 + iy) = i\sqrt{y^2 - 1}.$$

(iii)  $y < -1$ .  $\theta_1 = \theta_2 = -\pi/2 \pmod{2\pi}$ , so  $\phi(y) = -\pi/2$  and

$$f(\pm 0 + iy) = -i\sqrt{y^2 - 1}.$$

### Branch cut and behaviour at infinity

The only jump occurs for  $|y| < 1$  on the imaginary axis, hence the branch cut is

$$S = \{x + iy : x = 0, |y| \leq 1\}.$$

For  $|z| \rightarrow \infty$ ,

$$f(z) = |z|e^{i \arg z} (1 + o(1)) = z(1 + o(1)),$$

so  $f(z) \sim z$ . The image of  $S$  is the real segment  $[-1, 1]$ ; the imaginary axis outside  $S$  maps to the imaginary axis. Thus  $f : \mathbb{C} \setminus S \rightarrow \mathbb{C} \setminus [-1, 1]$  is conformal with  $f(z) \sim z$ .

### 4(c). A branch with two semi-infinite vertical cuts

Now define

$$f(z) = (r_1 r_2)^{1/2} e^{i(\theta_1 + \theta_2)/2}, \quad -\frac{3\pi}{2} < \theta_1 \leq \frac{\pi}{2}, \quad -\frac{\pi}{2} < \theta_2 \leq \frac{3\pi}{2}.$$

#### Values across the imaginary axis

Let  $z = x + iy$ ,  $x \rightarrow 0^\pm$ .

- $|y| < 1$ : we may keep  $\theta_1 \approx -\pi/2$ ,  $\theta_2 \approx \pi/2$  on both sides, hence

$$f(\pm 0 + iy) = \sqrt{1 - y^2} \quad (\text{no jump}).$$

- $y > 1$ : crossing the axis forces  $\theta_1$  to jump by  $2\pi$ ; thus

$$f(0^+ + iy) = +i\sqrt{y^2 - 1}, \quad f(0^- + iy) = -i\sqrt{y^2 - 1}.$$

- $y < -1$ : analogously,

$$f(0^+ + iy) = -i\sqrt{y^2 - 1}, \quad f(0^- + iy) = +i\sqrt{y^2 - 1}.$$

Therefore the branch cut consists of the two rays  $(-\infty i, -i] \cup [i, +\infty i)$ .

**Values on the real axis and asymptotics by half-planes**

For  $z = x \in \mathbb{R}$ ,

$$r_1 = r_2 = \sqrt{x^2 + 1}, \quad \frac{\theta_1 + \theta_2}{2} = \begin{cases} 0, & x > 0, \\ \pi, & x < 0. \end{cases}$$

Hence

$$\boxed{f(x) = \operatorname{sgn}(x) \sqrt{x^2 + 1}}.$$

As  $|z| \rightarrow \infty$ ,

$$f(z) \sim |z| e^{i \arg z} \times \begin{cases} 1, & \Re z > 0, \\ -1, & \Re z < 0, \end{cases}$$

i.e.

$$\boxed{f(z) \sim z \text{ for } \Re z > 0, \quad f(z) \sim -z \text{ for } \Re z < 0.}$$

**Exact problem statement (as in the text)****4. Branches of  $\sqrt{z^2 + 1}$  — Problem statement**

(a) **(a)** Show that, if  $\zeta^2 = z^2 + 1$  and

$$z = i + r_1 e^{i\theta_1} = -i + r_2 e^{i\theta_2}, \quad r_1, r_2 > 0, \quad \theta_1, \theta_2 \in \mathbb{R},$$

then

$$\zeta = \pm (r_1 r_2)^{1/2} e^{i(\theta_1 + \theta_2)/2}.$$

Explain briefly why  $z = \pm i$  are the branch points of the multifunction  $(z^2 + 1)^{1/2}$ .

(b) **(b)** Consider the branch defined by

$$f(z) = (r_1 r_2)^{1/2} e^{i(\theta_1 + \theta_2)/2}, \quad -\frac{\pi}{2} < \theta_1, \theta_2 \leq \frac{3\pi}{2}.$$

State the values of  $(\theta_1 + \theta_2)/2$  on either side of the imaginary axis, and hence compute  $f(\pm 0 + iy)$  in terms of  $y$  for  $y \in \mathbb{R}$ . Deduce that the branch cut is

$$S = \{x + iy : x = 0, |y| \leq 1\}.$$

Show that  $f(z) \sim z$  as  $|z| \rightarrow \infty$  (i.e.  $f(z)/z \rightarrow 1$  as  $|z| \rightarrow \infty$ ). Sketch the image of the cut  $z$ -plane  $\mathbb{C} \setminus S$  under the map  $\zeta = f(z)$ .

(c) **(c)** Consider instead the branch

$$f(z) = (r_1 r_2)^{1/2} e^{i(\theta_1 + \theta_2)/2}, \quad -\frac{3\pi}{2} < \theta_1 \leq \frac{\pi}{2}, \quad -\frac{\pi}{2} < \theta_2 \leq \frac{3\pi}{2}.$$

State the values of  $(\theta_1 + \theta_2)/2$  on either side of the imaginary axis, and hence compute  $f(\pm 0 + iy)$  in terms of  $y$  for  $y \in \mathbb{R}$ . Deduce that the branch cut is along the imaginary axis from  $z = -i\infty$  to  $z = -i$  and from  $z = i$  to  $z = i\infty$ . Compute  $f(x)$  in terms of  $x$  for  $x \in \mathbb{R}$ . Show that

$$f(z) \sim \begin{cases} z, & \text{as } |z| \rightarrow \infty \text{ with } \Re(z) > 0, \\ -z, & \text{as } |z| \rightarrow \infty \text{ with } \Re(z) < 0. \end{cases}$$

Sketch the images of the half-planes  $\Re(z) > 0$  and  $\Re(z) < 0$  under the map  $\zeta = f(z)$ .

## Solutions (with background)

Throughout, write

$$z - i = r_1 e^{i\theta_1}, \quad z + i = r_2 e^{i\theta_2}, \quad r_1, r_2 > 0,$$

so that

$$z^2 + 1 = (z - i)(z + i) = r_1 r_2 e^{i(\theta_1 + \theta_2)}.$$

### (a) Formula for $\zeta$ and branch points

Since  $\zeta^2 = r_1 r_2 e^{i(\theta_1 + \theta_2)}$ , any square root is

$$\boxed{\zeta = \pm (r_1 r_2)^{1/2} e^{i(\theta_1 + \theta_2)/2}}.$$

Branch points occur at zeros of  $z^2 + 1$ , where an argument for a factor cannot be chosen continuously: these are exactly  $z = \pm i$ .

### (b) Branch with finite vertical cut $S = \{x = 0, |y| \leq 1\}$

**Branch choice.** Take  $-\frac{\pi}{2} < \theta_1, \theta_2 \leq \frac{3\pi}{2}$  and define  $f(z) = (r_1 r_2)^{1/2} e^{i(\theta_1 + \theta_2)/2}$ .

**Angles on the imaginary axis.** Let  $z = x + iy$  and denote  $\phi(y) = \frac{1}{2}(\theta_1 + \theta_2)$ . Use

$$r_1 = |y - 1|, \quad r_2 = |y + 1|, \quad \arg(it) = \begin{cases} \pi/2, & t > 0, \\ -\pi/2, & t < 0. \end{cases}$$

(i)  $|y| < 1$ . As  $x \rightarrow 0^+$ :  $\theta_1 \rightarrow (-\pi/2)^+$ ,  $\theta_2 \rightarrow (\pi/2)^- \Rightarrow \phi(y) \rightarrow 0$ . As  $x \rightarrow 0^-$ : to keep  $\theta_1$  in  $(-\pi/2, 3\pi/2]$  we add  $2\pi$ , so  $\theta_1 \rightarrow (3\pi/2)^-$ ,  $\theta_2 \rightarrow (\pi/2)^+ \Rightarrow \phi(y) \rightarrow \pi$ . Hence

$$\boxed{f(0^+ + iy) = \sqrt{1 - y^2}, \quad f(0^- + iy) = -\sqrt{1 - y^2} \quad (|y| < 1)}.$$

(ii)  $y > 1$ .  $\theta_1 = \theta_2 = \pi/2$  from either side, so

$$\boxed{f(\pm 0 + iy) = i\sqrt{y^2 - 1} \quad (y > 1)}.$$

(iii)  $y < -1$ .  $\theta_1 = \theta_2 = -\pi/2 \pmod{2\pi}$ , so

$$\boxed{f(\pm 0 + iy) = -i\sqrt{y^2 - 1} \quad (y < -1)}.$$

**Branch cut.** The only sign jump occurs for  $|y| < 1$  on the imaginary axis, hence

$$\boxed{S = \{x + iy : x = 0, |y| \leq 1\}}.$$

**Asymptotics.** As  $|z| \rightarrow \infty$ ,

$$r_1 r_2 = |z - i||z + i| = |z^2 + 1| = |z|^2(1 + o(1)), \quad \theta_1 + \theta_2 = \arg(z^2 + 1) = 2 \arg z + o(1),$$

so

$$\boxed{f(z) \sim z \quad (|z| \rightarrow \infty)}.$$

**Image sketch.** The cut segment  $S$  maps to the real interval  $[-1, 1]$  (opposite sides map to  $(0, 1]$  and  $[-1, 0)$ ); outside  $S$  the imaginary axis maps to the imaginary axis. Thus  $f : \mathbb{C} \setminus S \rightarrow \mathbb{C} \setminus [-1, 1]$  is conformal with  $f(z) \sim z$ .

**(c) Branch with two semi-infinite cuts**

**Branch choice.** Take  $-\frac{3\pi}{2} < \theta_1 \leq \frac{\pi}{2}$  and  $-\frac{\pi}{2} < \theta_2 \leq \frac{3\pi}{2}$ , and define  $f(z) = (r_1 r_2)^{1/2} e^{i(\theta_1 + \theta_2)/2}$ .

**Angles across the imaginary axis.** For  $z = x + iy$ :

(i)  $|y| < 1$ . We may keep  $\theta_1 \approx -\pi/2$  and  $\theta_2 \approx \pi/2$  on both sides, hence

$$f(\pm 0 + iy) = \sqrt{1 - y^2} \quad (|y| < 1).$$

(ii)  $y > 1$ . Crossing the axis forces  $\theta_1$  to jump by  $2\pi$ ; therefore

$$f(0^+ + iy) = +i\sqrt{y^2 - 1}, \quad f(0^- + iy) = -i\sqrt{y^2 - 1} \quad (y > 1).$$

(iii)  $y < -1$ . Similarly,

$$f(0^+ + iy) = -i\sqrt{y^2 - 1}, \quad f(0^- + iy) = +i\sqrt{y^2 - 1} \quad (y < -1).$$

**Branch cuts.** The discontinuities occur on the rays

$$(-i\infty, -i] \cup [i, +i\infty).$$

**Values on the real axis.** For  $z = x \in \mathbb{R}$ ,  $r_1 = r_2 = \sqrt{x^2 + 1}$  and

$$\frac{\theta_1 + \theta_2}{2} = \begin{cases} 0, & x > 0, \\ \pi, & x < 0. \end{cases} \Rightarrow \boxed{f(x) = \operatorname{sgn}(x) \sqrt{x^2 + 1}}.$$

**Asymptotics by half-planes.** As  $|z| \rightarrow \infty$ ,

$$\boxed{f(z) \sim z \text{ for } \Re z > 0, \quad f(z) \sim -z \text{ for } \Re z < 0}.$$

## 4 Problem 5: Images under $\zeta = \frac{1}{z}$ , $\zeta = \sin z$ , and $\zeta = z + e^z$

**(a) Intersection of  $|z - 1| < 1$  and  $|z + i| < 1$  under  $\zeta = 1/z$**

**Background (inversion).** If  $w = 1/z$  and  $|z - a| = |a|$  (a circle through 0), then

$$|1 - aw| = |a||w| \iff 1 = 2\Re(aw) \iff \Re(aw) = \frac{1}{2},$$

i.e. the image is a straight line.

**Boundary images.**

$$|z - 1| = 1 \mapsto \Re(w) = \frac{1}{2}, \quad |z + i| = 1 \mapsto \Re((-i)w) = \frac{1}{2} \iff \Im(w) = \frac{1}{2}.$$

**Which side is the interior?**

$$|z - 1| < 1 \iff |1/w - 1| < 1 \iff |1 - w| < |w| \iff \Re(w) > \frac{1}{2};$$

$$|z + i| < 1 \iff |1 + iw| < |w| \iff \Im(w) > \frac{1}{2}.$$

**Image.** The lens  $|z - 1| < 1 \cap |z + i| < 1$  maps to the quarter-plane

$$\boxed{\{w : \Re w > \frac{1}{2}, \Im w > \frac{1}{2}\}}.$$


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**(b) Strip**  $-\frac{\pi}{2} < x < \frac{\pi}{2}$  **under**  $\zeta = \sin z$

Write  $z = x + iy$ . Then

$$\sin z = \sin x \cosh y + i \cos x \sinh y.$$

**Boundary lines.**

$$\sin\left(\pm \frac{\pi}{2} + iy\right) = \pm \cosh y \in \mathbb{R}, \quad |\cosh y| \geq 1.$$

Hence the boundaries map onto the real rays  $[1, \infty)$  and  $(-\infty, -1]$ .

**Conformality.**  $\cos z \neq 0$  on  $|x| < \pi/2$  (zeros lie on  $x = \pi/2 + k\pi$ ), so  $\sin$  is conformal and injective on the strip.

**Image.** The image is the simply connected domain whose boundary is those rays; by symmetry (the line  $x = 0$  maps to the entire imaginary axis),

$$\boxed{\sin : \{-\frac{\pi}{2} < \Re z < \frac{\pi}{2}\} \xrightarrow{1-1} \mathbb{C} \setminus ((-\infty, -1] \cup [1, \infty)).}$$


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**(c) Strip**  $-\pi < y < \pi$  **under**  $\zeta = f(z) = z + e^z$

Let  $z = x + iy$ . Then

$$f(z) = (x + e^x \cos y) + i(y + e^x \sin y).$$

**Critical points (hint).**

$$f'(z) = 1 + e^z = 0 \iff e^z = -1 \iff z = i(2k + 1)\pi.$$

There are no critical points in the open strip  $-\pi < y < \pi$ ; the boundary points  $z = \pm i\pi$  are the only adjacent critical points.

**Boundary images.** For  $y = \pm\pi$ ,  $\sin y = 0$ ,  $\cos y = -1$ :

$$f(x \pm i\pi) = (x - e^x) \pm i\pi.$$

Here  $u(x) = x - e^x$  has a unique maximum  $-1$  at  $x = 0$  and  $u(x) \rightarrow -\infty$  as  $x \rightarrow \pm\infty$ . Thus each boundary maps to a horizontal half-line

$$(-\infty, -1] + i\pi, \quad (-\infty, -1] - i\pi,$$

meeting at the critical values  $-1 \pm i\pi$ .

**Interior and image.** With no interior critical points,  $f$  is conformal on the strip. For fixed  $y \in (-\pi, \pi)$ ,  $f(x + iy) \rightarrow -\infty + iy$  as  $x \rightarrow -\infty$  and  $f(x + iy) \sim e^x e^{iy} \rightarrow \infty$  along the ray  $\arg \zeta = y$  as  $x \rightarrow +\infty$ . Hence the image is the plane with the two horizontal half-lines removed:

$$\mathbb{C} \setminus \left( (-\infty, -1] + i\pi \cup (-\infty, -1] - i\pi \right).$$

## 5 Problem 6 — Conformal mappings

### Exact problem statement

(a)  $D$  is the region exterior to the two circles  $|z - 1| = 1$  and  $|z + 1| = 1$ . Find a conformal mapping from  $D$  to the exterior of the unit circle.

*Hint:* First apply an inversion with respect to the origin ( $z \mapsto 1/z$ ), then rotate and scale so as to use the exponential function to get to a half plane, from where use Möbius.

(b) Find a conformal map from the unit disc  $|z| < 1$  onto the strip  $-\pi/2 < \eta < \pi/2$ , taking the origin to the origin, 1 to  $\xi = +\infty$ ,  $-1$  to  $\xi = -\infty$ . (Thus, the upper half of the unit circle maps to  $\eta = \pi/2$  and the lower half to  $\eta = -\pi/2$ .)

(c) Find a conformal map of the quarter-disc  $0 < |z| < 1$ ,  $0 < \arg z < \pi/2$  to the upper half-plane  $\eta > 0$ , taking 0 to 0, 1 to 1, and  $i$  to  $\infty$ .

### Solutions (step by step)

(a) **Exterior of the two circles  $\rightarrow$  exterior of  $|\zeta| = 1$ .**

- **Invert.** Let  $w = 1/z$ . Then the circles through 0 become lines:

$$|z - 1| = 1 \Rightarrow \Re w = \frac{1}{2}, \quad |z + 1| = 1 \Rightarrow \Re w = -\frac{1}{2}.$$

Hence  $D$  maps to the vertical strip  $S = \{-\frac{1}{2} < \Re w < \frac{1}{2}\}$ .

- **Rotate/scale to a horizontal strip.** Put  $s = i\pi w$ . Then  $|\Im s| < \pi/2$ .
- **Exponential to a half-plane.** Let  $u = e^s$ . Then  $|\Im s| < \pi/2 \implies \Re u > 0$ .
- **Möbius to exterior unit disk.** Use

$$\Phi(u) = \frac{u + 1}{u - 1}, \quad \Re u > 0 \mapsto |\zeta| > 1.$$

Therefore a concrete map is

$$\zeta = \Phi(e^{i\pi/z}) = \frac{e^{i\pi/z} + 1}{e^{i\pi/z} - 1}.$$

(b) **Unit disk**  $\rightarrow$  **strip**  $-\pi/2 < \Im \zeta < \pi/2$ .

- **Disk to right half-plane (Cayley).**

$$w = \frac{1+z}{1-z}, \quad |z| < 1 \mapsto \Re w > 0,$$

and  $z = 0 \mapsto w = 1$ ,  $z = 1 \mapsto w = \infty$ ,  $z = -1 \mapsto w = 0$ .

- **Log to a horizontal strip.**

$$\zeta = \log w \quad (\text{principal branch}), \quad \Im \zeta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right).$$

Then  $\zeta(0) = 0$ ,  $\zeta(1) = +\infty$ ,  $\zeta(-1) = -\infty$ .

Hence

$$\boxed{\zeta = \log \left( \frac{1+z}{1-z} \right)}.$$

(c) **Quarter-disk**  $\rightarrow$  **upper half-plane**;  $0 \mapsto 0$ ,  $1 \mapsto 1$ ,  $i \mapsto \infty$ .

- **Square to upper semicircle.**

$$t = z^2 : \quad \{0 < |z| < 1, 0 < \arg z < \frac{\pi}{2}\} \mapsto \{|t| < 1, \Im t > 0\}.$$

Moreover  $0 \mapsto 0$ ,  $1 \mapsto 1$ ,  $i \mapsto -1$ .

- **Real Möbius to upper half-plane.** Find  $M(t)$  with  $M(-1) = \infty$ ,  $M(0) = 0$ ,  $M(1) = 1$ .  
A suitable choice is

$$M(t) = \frac{2t}{t+1}.$$

Therefore

$$\boxed{\zeta = M(z^2) = \frac{2z^2}{z^2+1}}.$$

It maps the quarter-disk conformally onto  $\Im \zeta > 0$  and has  $\zeta(0) = 0$ ,  $\zeta(1) = 1$ ,  $\zeta(i) = \infty$ .

**6(c) Quarter-disk**  $0 < |z| < 1$ ,  $0 < \arg z < \frac{\pi}{2} \rightarrow$  **upper half-plane**  
 $\Im \zeta > 0$ , **with**  $0 \mapsto 0$ ,  $1 \mapsto 1$ ,  $i \mapsto \infty$

**Goal**

Construct an explicit conformal map  $\zeta = \zeta(z)$  that maps the open quarter-disk

$$Q = \{z : 0 < |z| < 1, 0 < \arg z < \frac{\pi}{2}\}$$

onto the upper half-plane  $\mathbb{H} = \{\zeta : \Im \zeta > 0\}$ , sending the three marked boundary points

$$z = 0 \mapsto \zeta = 0, \quad z = 1 \mapsto \zeta = 1, \quad z = i \mapsto \zeta = \infty.$$

**Step 1: Square to unfold the angle**

Let

$$t = z^2.$$

Write  $z = re^{i\theta}$  with  $0 < r < 1$  and  $0 < \theta < \frac{\pi}{2}$ . Then

$$t = r^2 e^{i(2\theta)} \Rightarrow |t| < 1, \quad 0 < \arg t < \pi \iff \Im t > 0.$$

Therefore the squaring map  $z \mapsto t = z^2$  sends  $Q$  *bijectively and conformally* onto the *open upper semicircle*

$$S = \{t : |t| < 1, \Im t > 0\}.$$

On the boundary rays/arcs we have

$$0 \mapsto 0, \quad 1 \mapsto 1, \quad i \mapsto -1.$$

(Conformality is immediate, since  $(z^2)' = 2z \neq 0$  on  $Q$ .)

**Step 2: Real Möbius map fixing the three marked points and sending  $S$  to  $\mathbb{H}$** 

Seek a real-coefficient Möbius transformation

$$M(t) = \frac{at + b}{ct + d}, \quad a, b, c, d \in \mathbb{R}, \quad ad - bc \neq 0,$$

such that

$$M(-1) = \infty, \quad M(0) = 0, \quad M(1) = 1.$$

Imposing the conditions:

- $M(-1) = \infty \Rightarrow c(-1) + d = 0 \Rightarrow d = c.$
- $M(0) = 0 \Rightarrow b/d = 0 \Rightarrow b = 0.$
- $M(1) = 1 \Rightarrow a/(c + d) = 1 \Rightarrow a = c + d = 2c.$

Up to an overall nonzero constant (which does not change the map), we can take  $c = 1$  and obtain

$$\boxed{M(t) = \frac{2t}{t + 1}}.$$

**Why  $M$  maps  $S$  into  $\Im \zeta > 0$  (and in fact maps the whole upper half-plane to itself).**

Let  $t = x + iy$  with  $y > 0$ . Then

$$M(t) = \frac{2t}{1+t}, \quad \Im M(t) = \frac{2\Im(t(1+\bar{t}))}{|1+t|^2} = \frac{2\Im t}{|1+t|^2} = \frac{2y}{|1+t|^2} > 0.$$

Hence  $M$  maps  $\{\Im t > 0\}$  into itself; in particular it maps the upper semicircle  $S$  (which lies in  $\Im t > 0$ ) into the upper half-plane. Moreover

$$M'(t) = \frac{2}{(1+t)^2} \neq 0 \quad \text{on } S,$$

so  $M$  is conformal (locally injective) there. The three-point normalization guarantees that it sends the three distinguished boundary points as prescribed:

$$M(-1) = \infty, \quad M(0) = 0, \quad M(1) = 1.$$

**Step 3: Compose and verify**

Define the desired map

$$\zeta(z) = M(z^2) = \frac{2z^2}{z^2 + 1}.$$

Then the normalization follows immediately:

$$\zeta(0) = 0, \quad \zeta(1) = 1, \quad \zeta(i) = \infty.$$

Because  $t = z^2$  has  $\Im t > 0$  on  $Q$  and  $\Im M(t) > 0$  for  $\Im t > 0$ , we have

$$\Im \zeta(z) = \Im(M(z^2)) > 0 \quad \text{for all } z \in Q.$$

Also  $\zeta'(z) = M'(z^2) \cdot 2z = \frac{4z}{(1+z^2)^2} \neq 0$  on  $Q$ , hence the map is conformal throughout the domain.

**Boundary correspondence (for intuition)**

- The *real radius* segment  $\{0 < z < 1\}$  maps by  $t = z^2$  to  $(0, 1)$  and then by  $M$  to  $(0, 1) \subset \mathbb{R}$ ; thus that edge lands on the real axis between 0 and 1.
- The *imaginary radius* segment  $\{0 < iz < i\}$  maps by  $t = z^2$  to  $(0, -1)$  and then by  $M$  to  $(0, \infty)$  on  $\mathbb{R}$ ; thus the edge  $0 \rightarrow i$  lands on the ray  $[0, \infty)$ .
- The *circular arc*  $\{e^{i\theta} : 0 < \theta < \frac{\pi}{2}\}$  maps by  $t = z^2$  to the unit semicircle  $\{e^{i\phi} : 0 < \phi < \pi\}$  and then by  $M$  to the real line as a boundary set.

The interior of the quarter-disk maps onto  $\Im \zeta > 0$  by the positivity computation for  $\Im M(t)$ .

**Conclusion**

The composition

$$\zeta(z) = \frac{2z^2}{z^2 + 1}$$

is a conformal bijection from the open quarter-disk  $Q$  onto the upper half-plane  $\mathbb{H}$ , satisfying the three mapping conditions  $0 \mapsto 0$ ,  $1 \mapsto 1$ ,  $i \mapsto \infty$ .

**(b) Unit disk  $\rightarrow$  strip  $-\frac{\pi}{2} < \Im \zeta < \frac{\pi}{2}$**

**Disk to right half-plane (Cayley).**

$$w = \frac{1+z}{1-z}, \quad |z| < 1 \implies \operatorname{Re} w > 0,$$

and on the marked points

$$z = 0 \mapsto w = 1, \quad z = 1 \mapsto w = \infty, \quad z = -1 \mapsto w = 0.$$

Moreover, the unit circle  $|z| = 1 \setminus \{1\}$  maps to the imaginary axis  $\operatorname{Re} w = 0$ .

**Log to a horizontal strip.** Use the principal logarithm on the right half-plane:

$$\zeta = \log w = \ln |w| + i \arg(w), \quad \arg(w) \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right),$$

so

$$-\frac{\pi}{2} < \Im \zeta < \frac{\pi}{2}.$$

At the marked points:

$$\zeta(0) = \log 1 = 0, \quad \zeta(1) = \log(\infty) = +\infty, \quad \zeta(-1) = \log(0) = -\infty.$$

**Which boundary goes where.** For  $z = e^{i\theta}$ ,

$$w = \frac{1 + e^{i\theta}}{1 - e^{i\theta}} = -i \cot\left(\frac{\theta}{2}\right) \in i\mathbb{R}.$$

Hence

$$0 < \theta < \pi \text{ (upper semicircle)} \Rightarrow \Im w < 0 \Rightarrow \Im \zeta = -\frac{\pi}{2},$$

$$\pi < \theta < 2\pi \text{ (lower semicircle)} \Rightarrow \Im w > 0 \Rightarrow \Im \zeta = +\frac{\pi}{2}.$$

(If one wants the upper semicircle to land on  $\Im \zeta = +\pi/2$ , use  $\zeta = -\log\left(\frac{1+z}{1-z}\right)$ .)

**Hence**

$$\boxed{\zeta = \log\left(\frac{1+z}{1-z}\right)}$$

is a conformal bijection from  $|z| < 1$  onto the strip  $-\pi/2 < \Im \zeta < \pi/2$  with  $0 \mapsto 0$ ,  $1 \mapsto +\infty$ ,  $-1 \mapsto -\infty$ .

**(a) Exterior of the two circles  $|z - 1| = 1$  and  $|z + 1| = 1$   $\rightarrow$  exterior of  $|\zeta| = 1$**

**Goal.** Let

$$D = \{z \in \mathbb{C} : |z - 1| \geq 1, |z + 1| \geq 1\} \setminus (\{|z - 1| = 1\} \cup \{|z + 1| = 1\}),$$

the region *exterior to both* unit circles centered at  $\pm 1$ . Construct an explicit conformal bijection  $F : D \rightarrow \{\zeta : |\zeta| > 1\}$ .

**Step 1: Inversion.** Let  $w = \frac{1}{z}$ . For any  $a \in \mathbb{C}$ , the circle  $\{|z - a| = |a|\}$  (which passes through 0) satisfies

$$|z - a| = |a| \iff \left|\frac{1}{w} - a\right| = |a| \iff |1 - aw| = |a||w| \iff 1 = 2\Re(aw).$$

Hence circles through 0 invert to *lines*. For  $a = 1$  and  $a = -1$  we obtain

$$|z - 1| = 1 \iff \Re w = \frac{1}{2}, \quad |z + 1| = 1 \iff \Re w = -\frac{1}{2}.$$

Which side is the exterior? Using  $|1 - aw|^2 = (1 - 2\Re(aw) + |a|^2|w|^2)$  we compute

$$|z - 1| > 1 \iff |1 - w| > |w| \iff 1 - 2\Re w > 0 \iff \Re w < \frac{1}{2},$$

$$|z + 1| > 1 \iff |1 + w| > |w| \iff 1 + 2\Re w > 0 \iff \Re w > -\frac{1}{2}.$$

Therefore the inversion  $w = 1/z$  sends  $D$  bijectively onto the *vertical strip*

$$S = \left\{ w \in \mathbb{C} : -\frac{1}{2} < \Re w < \frac{1}{2} \right\}.$$

(Since  $z \neq 0$  on  $D$ , the inversion is analytic and injective there.)

**Step 2: Rotate/scale to a horizontal strip.** Define

$$s = i\pi w.$$

Write  $w = u + iv$  with  $-\frac{1}{2} < u < \frac{1}{2}$ . Then  $s = i\pi(u + iv) = -\pi v + i\pi u$ , so

$$|\Im s| = |\pi u| < \frac{\pi}{2}.$$

Consequently  $S$  is mapped conformally onto the horizontal strip

$$T = \left\{ s \in \mathbb{C} : |\Im s| < \frac{\pi}{2} \right\}.$$

**Step 3: Exponential to the right half-plane.** Set

$$u = e^s.$$

If  $s = \sigma + i\tau$  with  $|\tau| < \pi/2$ , then

$$\Re u = \Re(e^\sigma(\cos \tau + i \sin \tau)) = e^\sigma \cos \tau > 0,$$

because  $\cos \tau > 0$  on  $(-\pi/2, \pi/2)$ . Hence  $e^{(\cdot)}$  maps  $T$  conformally and bijectively onto the *right half-plane*

$$H = \{ u \in \mathbb{C} : \Re u > 0 \},$$

and the boundary lines  $\Im s = \pm\pi/2$  go to the imaginary axis  $\Re u = 0$ .

**Step 4: Möbius map from the right half-plane to the exterior disk.** Consider

$$\Phi(u) = \frac{u + 1}{u - 1}.$$

For  $u \in H$  we have the equivalence

$$|\Phi(u)| > 1 \iff |u + 1| > |u - 1| \iff (u + 1)(\bar{u} + 1) > (u - 1)(\bar{u} - 1) \iff 4\Re u > 0,$$

which holds precisely on  $H$ . Thus  $\Phi$  maps  $H$  conformally onto  $\{\zeta : |\zeta| > 1\}$  and sends the imaginary axis to the unit circle  $|\zeta| = 1$ .

**Step 5: Composition and explicit formula.** The composition of the four conformal bijections yields

$$F : D \xrightarrow{z \mapsto w=1/z} S \xrightarrow{w \mapsto s=i\pi w} T \xrightarrow{s \mapsto u=e^s} H \xrightarrow{u \mapsto \Phi(u)} \{|\zeta| > 1\}.$$

Therefore an explicit mapping is

$$\boxed{\zeta = F(z) = \Phi(e^{i\pi/z}) = \frac{e^{i\pi/z} + 1}{e^{i\pi/z} - 1}}.$$

Every factor is analytic and injective on the relevant domain; hence  $F$  is conformal and bijective from  $D$  onto  $\{|\zeta| > 1\}$ .

**Asymptotic check at infinity (sanity).** As  $z \rightarrow \infty$  we have  $e^{i\pi/z} = 1 + \frac{i\pi}{z} + O(z^{-2})$ , so

$$\zeta = \frac{2 + \frac{i\pi}{z} + O(z^{-2})}{\frac{i\pi}{z} + O(z^{-2})} = \frac{2}{i\pi} z \left(1 + O(z^{-1})\right) = -\frac{2i}{\pi} z \left(1 + O(z^{-1})\right),$$

which confirms that far out in  $D$  the map behaves like an affine scaling of  $z$ , as expected.

**Boundary correspondence (orientation).**

- The line  $\Re w = \frac{1}{2}$  (image of  $|z - 1| = 1$ ) becomes  $\Im s = \frac{\pi}{2}$ , then the imaginary axis, and finally the unit circle  $|\zeta| = 1$ .
- The line  $\Re w = -\frac{1}{2}$  (image of  $|z + 1| = 1$ ) becomes  $\Im s = -\frac{\pi}{2}$ , then the same imaginary axis, and again  $|\zeta| = 1$ .
- The interior  $-\frac{1}{2} < \Re w < \frac{1}{2}$  maps to  $|\Im s| < \frac{\pi}{2}$ , then to  $\Re u > 0$ , and finally to  $|\zeta| > 1$ .

This completes the construction and detailed verification.

## 6 Problem 7: Temperature in a right half-plane with a removed disk

**Exact statement (verbatim)**

The domain  $D$  consists of the right-hand half plane  $x > 0$  with the circle  $|z - a| = b$ ,  $0 < b < a$ , and its interior removed. Find the temperature  $u(x, y)$  in steady heat flow if  $u = 0$  on the  $y$  axis,  $u = 1$  on  $|z - a| = b$ , and  $u \rightarrow 0$  at infinity.

*Hint:* Show that the mapping  $\zeta = \frac{z - \alpha}{z + \alpha}$ , with  $\alpha$  real and positive, takes  $D$  onto an annular region with the imaginary axis mapping to  $|\zeta| = 1$  and show that, if  $\alpha^2 = a^2 - b^2$ , then the image of  $D$  is a concentric circular annulus.

**Background: why the suggested Möbius map produces an annulus**

Consider the real-parameter Möbius map

$$\zeta = \zeta(z) = \frac{z - \alpha}{z + \alpha}, \quad \alpha > 0.$$

**Imaginary axis.** For  $z = iy$  we have

$$\left| \frac{iy - \alpha}{iy + \alpha} \right| = 1 \quad (\text{numerator and denominator are conjugates}),$$

so the  $y$ -axis maps to the unit circle  $|\zeta| = 1$ .

**Image of the circle  $|z - a| = b$ .** Write  $z = a + be^{i\theta}$  and compute

$$|\zeta|^2 = \frac{|a - \alpha + be^{i\theta}|^2}{|a + \alpha + be^{i\theta}|^2} = \frac{A + B \cos \theta}{C + D \cos \theta},$$

where

$$A = (a - \alpha)^2 + b^2, \quad B = 2b(a - \alpha), \quad C = (a + \alpha)^2 + b^2, \quad D = 2b(a + \alpha).$$

For  $|\zeta|$  to be constant on the circle (i.e. the image is a circle *centered at the origin*), the  $\cos \theta$ -dependence must drop out:

$$\frac{A + B \cos \theta}{C + D \cos \theta} \equiv \frac{A}{C} \iff AD = BC.$$

We now verify that

$$AD - BC = 2b \left[ (a + \alpha)((a - \alpha)^2 + b^2) - (a - \alpha)((a + \alpha)^2 + b^2) \right] = 4b\alpha(\alpha^2 + b^2 - a^2).$$

Thus  $AD = BC$  is equivalent to

$$\boxed{\alpha^2 = a^2 - b^2}.$$

With this choice (note  $a > b > 0 \Rightarrow \alpha > 0$ ), the ratio becomes *constant*:

$$|\zeta|^2 = \frac{A}{C} = \frac{(a - \alpha)^2 + b^2}{(a + \alpha)^2 + b^2} = \frac{2a(a - \alpha)}{2a(a + \alpha)} = \frac{a - \alpha}{a + \alpha}.$$

Hence the circle  $|z - a| = b$  maps to the circle  $|\zeta| = \rho$  with

$$\boxed{\rho = \sqrt{\frac{a - \alpha}{a + \alpha}} \in (0, 1)}.$$

**Image of the domain.** Therefore, for  $\alpha = \sqrt{a^2 - b^2}$ , the map  $z \mapsto \zeta$  sends the domain  $D$  (right half-plane with the closed disk  $\{|z - a| \leq b\}$  removed) onto the concentric circular *annulus*

$$\mathcal{A} = \{ \rho < |\zeta| < 1 \}.$$

The boundary pieces correspond as follows:

$$\{x = 0\} \mapsto |\zeta| = 1, \quad \{|z - a| = b\} \mapsto |\zeta| = \rho.$$

## Solving the Dirichlet problem on the annulus

Let  $U(\zeta)$  denote the temperature in the  $\zeta$ -plane. Harmonicity is preserved by conformal maps, so  $U$  is harmonic on  $\mathcal{A}$  and the boundary data become

$$U = 0 \quad \text{on} \quad |\zeta| = 1, \quad U = 1 \quad \text{on} \quad |\zeta| = \rho.$$

By rotational symmetry,  $U$  depends only on  $r = |\zeta|$ . The general radial harmonic function on an annulus is  $A \log r + B$ . Imposing the boundary values gives

$$U(r) = \frac{\log(1/r)}{\log(1/\rho)} = \frac{-\log r}{-\log \rho} = \frac{\log r}{\log \rho} \quad \text{for} \quad \rho < r < 1.$$

(Any of the three displayed forms is correct; we will use the first one momentarily.)

---

## Pullback to the original $z$ -plane and final formula

The physical temperature is  $u(z) = U(\zeta(z))$  with  $r = |\zeta(z)| = \left| \frac{z-\alpha}{z+\alpha} \right|$ . Hence

$$u(z) = U\left(\left|\frac{z-\alpha}{z+\alpha}\right|\right) = \frac{\log\left(\frac{1}{\left|\frac{z-\alpha}{z+\alpha}\right|}\right)}{\log\left(\frac{1}{\rho}\right)} = \frac{\log\left|\frac{z+\alpha}{z-\alpha}\right|}{\log\left(\frac{1}{\rho}\right)}.$$

Since  $\rho = \sqrt{\frac{a-\alpha}{a+\alpha}}$  we have

$$\log\left(\frac{1}{\rho}\right) = \frac{1}{2} \log\left(\frac{a+\alpha}{a-\alpha}\right).$$

Therefore a convenient explicit form is

$$u(z) = \frac{2 \log\left|\frac{z+\alpha}{z-\alpha}\right|}{\log\left(\frac{a+\alpha}{a-\alpha}\right)}, \quad \alpha = \sqrt{a^2 - b^2}.$$

## Checks of the boundary conditions and far field

- **On the  $y$ -axis  $x = 0$ :** for  $z = iy$  we have  $\left|\frac{z-\alpha}{z+\alpha}\right| = 1$ , so the numerator  $\log\left|\frac{z+\alpha}{z-\alpha}\right| = 0$  and hence  $u = 0$ .
  - **On the circle  $|z - a| = b$ :** by construction  $|\zeta| = \rho$ , i.e.  $\left|\frac{z-\alpha}{z+\alpha}\right| = \rho$ , so  $\log\left|\frac{z+\alpha}{z-\alpha}\right| = \log\left(\frac{1}{\rho}\right)$  and the fraction evaluates to  $u = 1$ .
  - **At infinity:** as  $|z| \rightarrow \infty$ ,  $\left|\frac{z+\alpha}{z-\alpha}\right| \rightarrow 1$ , so the numerator  $\rightarrow 0$  and  $u(z) \rightarrow 0$ .
-

## Summary

With  $\alpha = \sqrt{a^2 - b^2}$  and  $\rho = \sqrt{\frac{a - \alpha}{a + \alpha}}$ , the Möbius map  $\zeta = \frac{z - \alpha}{z + \alpha}$  carries  $D$  onto the annulus  $\{\rho < |\zeta| < 1\}$ . The unique harmonic function with  $U = 1$  on  $|\zeta| = \rho$  and  $U = 0$  on  $|\zeta| = 1$  is  $U(r) = \log(1/r)/\log(1/\rho)$ , and hence

$$u(z) = \frac{2 \log \left| \frac{z + \alpha}{z - \alpha} \right|}{\log \left( \frac{a + \alpha}{a - \alpha} \right)}.$$

This  $u$  satisfies all the prescribed boundary conditions and  $u(z) \rightarrow 0$  as  $|z| \rightarrow \infty$ .

## Hyperbolic maps: $\sinh$ , $\cosh$ , $\tanh$ , $\coth$ — background, zeros/poles, and strip mappings

### 1. $w = \sinh z$

**Background:** entire, odd; period  $2\pi i$ . **Zeros:**  $z = i\pi k$ ; **poles:** none.

**Decomposition:**  $w = u + iv = \sinh x \cos y + i \cosh x \sin y$ .

**Vertical lines**  $x = \text{const}$ :  $\left(\frac{u}{\sinh x}\right)^2 + \left(\frac{v}{\cosh x}\right)^2 = 1$  (ellipses, foci  $\pm i$ ).

**Horizontal lines**  $y = \text{const}$ :  $\left(\frac{u}{\cos y}\right)^2 - \left(\frac{v}{\sin y}\right)^2 = -1$  (hyperbolas, foci  $\pm i$ ).

**Strip mapping:**

$$\sinh : \{|\Im z| < \frac{\pi}{2}\} \xrightarrow{1-1} \mathbb{C} \setminus i((-\infty, -1] \cup [1, \infty))$$

Boundaries  $y = \pm \frac{\pi}{2}$  map to imaginary rays  $\pm i[1, \infty)$ .

### 2. $w = \cosh z$

**Background:** entire, even; period  $2\pi i$ . **Zeros:**  $z = i\pi(k + \frac{1}{2})$ ; **poles:** none.

**Decomposition:**  $w = u + iv = \cosh x \cos y + i \sinh x \sin y$ .

**Vertical lines**  $x = \text{const}$ :  $\left(\frac{u}{\cosh x}\right)^2 + \left(\frac{v}{\sinh x}\right)^2 = 1$  (ellipses, foci  $\pm 1$ ).

**Horizontal lines**  $y = \text{const}$ :  $\left(\frac{u}{\cos y}\right)^2 - \left(\frac{v}{\sin y}\right)^2 = 1$  (hyperbolas, foci  $\pm 1$ ).

**Strip mapping:**

$$\cosh : \{|\Im z| < \pi\} \xrightarrow{2-1} \mathbb{C} \setminus ((-\infty, -1] \cup [1, \infty))$$

On the half-strip  $0 < |\Im z| < \pi$ , the map is 1-1 onto the slit plane.

**3.**  $w = \tanh z$ **Background:** meromorphic, odd; period  $i\pi$ .    **Zeros:**  $z = i\pi k$ ; **poles:**  $z = i\pi(k + \frac{1}{2})$ .**Strip  $\rightarrow$  disk:**

$$\tanh : \{|\Im z| < \frac{\pi}{2}\} \xrightarrow{1-1} \mathbb{D}$$

Real line maps to  $(-1, 1)$ ; as  $|\Im z| \rightarrow \pi/2$ ,  $|\tanh z| \rightarrow 1$ .**4.**  $w = \coth z$ **Background:** meromorphic, odd; period  $i\pi$ .    **Zeros:**  $z = i\pi(k + \frac{1}{2})$ ; **poles:**  $z = i\pi k$ .**Strip  $\rightarrow$  exterior disk:**

$$\coth : \{|\Im z| < \frac{\pi}{2}\} \setminus \{0\} \xrightarrow{1-1} \{|w| > 1\}$$

Real line maps to  $(-\infty, -1) \cup (1, \infty)$ ;  $|\Im z| \rightarrow \pi/2$  gives  $|w| \rightarrow 1^+$ .**Hyperbolic Conformal Maps:**  $\sinh$ ,  $\cosh$ ,  $\tanh$ ,  $\coth$ **Notation and preliminaries**Throughout, write  $z = x + iy$ ,  $x, y \in \mathbb{R}$ , and  $w = u + iv$  for the image. Recall the basic identities

$$\cosh z = \frac{e^z + e^{-z}}{2}, \quad \sinh z = \frac{e^z - e^{-z}}{2}, \quad \tanh z = \frac{\sinh z}{\cosh z}, \quad \coth z = \frac{\cosh z}{\sinh z}.$$

We use the principal branch of the logarithm  $\log$  on  $\mathbb{C} \setminus (-\infty, 0]$ , and principal square roots  $\sqrt{\cdot}$  with branch cut on  $(-\infty, 0]$  unless stated otherwise.**1.**  $w = \sinh z$ **Analytic background.** $\sinh$  is an *entire*, *odd* function, with imaginary-periodicity:

$$\sinh(-z) = -\sinh z, \quad \sinh(z + 2\pi i) = \sinh z.$$

Zeros occur at  $z_k = i\pi k$  for  $k \in \mathbb{Z}$  (all simple). No poles.**Cauchy–Riemann decomposition.**With  $z = x + iy$ ,

$$\sinh z = \sinh x \cos y + i \cosh x \sin y \quad \Rightarrow \quad u = \sinh x \cos y, \quad v = \cosh x \sin y.$$

From this, one obtains orthogonal families of confocal conics:

$$\begin{aligned} x = \text{const} : \quad & \left( \frac{u}{\sinh x} \right)^2 + \left( \frac{v}{\cosh x} \right)^2 = 1 && (\text{ellipses, foci at } \pm i), \\ y = \text{const} : \quad & \left( \frac{u}{\cos y} \right)^2 - \left( \frac{v}{\sin y} \right)^2 = -1 && (\text{hyperbolas, foci at } \pm i). \end{aligned}$$

Real and imaginary axes map as:

$$y = 0 : w = \sinh x \in \mathbb{R} \text{ (onto } \mathbb{R}), \quad x = 0 : w = i \sin y \text{ (onto the segment } i[-1, 1]).$$

### Jacobian and conformality.

$\sinh'(z) = \cosh z$ , and  $\cosh z \neq 0$  on the strip  $|\Im z| < \frac{\pi}{2}$ ; hence  $\sinh$  is conformal there.

### Canonical strip mapping (global picture).

$$\sinh : \{ |\Im z| < \frac{\pi}{2} \} \xrightarrow{\text{biholomorphism}} \mathbb{C} \setminus i((-\infty, -1] \cup [1, \infty)).$$

*Explanation.* The boundary lines  $y = \pm \frac{\pi}{2}$  map to the imaginary rays  $\pm i[1, \infty)$  (via  $v = \cosh x \sin y$  with  $\sin(\pm\pi/2) = \pm 1$  and  $u = \sinh x \cos(\pm\pi/2) = 0$ ), while the interior maps injectively since  $\cosh z \neq 0$  and periodicity in the imaginary direction has period  $2\pi$ , not  $\pi$ .

### Inverse branch and branch cuts.

A principal inverse on the slit domain is

$$\operatorname{arsinh} w = \log(w + \sqrt{w^2 + 1}),$$

with branch cuts along  $i(-\infty, -1] \cup i[1, \infty)$  so that  $\Im(\operatorname{arsinh} w) \in (-\frac{\pi}{2}, \frac{\pi}{2})$ .

### Useful identities.

$$\cosh^2 z - \sinh^2 z = 1, \quad \sinh(z + \zeta) = \sinh z \cosh \zeta + \cosh z \sinh \zeta.$$

## 2. $w = \cosh z$

### Analytic background.

$\cosh$  is *entire*, *even*, and  $2\pi i$ -periodic:

$$\cosh(-z) = \cosh z, \quad \cosh(z + 2\pi i) = \cosh z.$$

Zeros at  $z = i\pi(k + \frac{1}{2})$ , all simple. No poles.

**Cauchy–Riemann decomposition.**

With  $z = x + iy$ ,

$$\cosh z = \cosh x \cos y + i \sinh x \sin y \quad \Rightarrow \quad u = \cosh x \cos y, \quad v = \sinh x \sin y.$$

Again one gets confocal conics:

$$\begin{aligned} x = \text{const} : \quad & \left( \frac{u}{\cosh x} \right)^2 + \left( \frac{v}{\sinh x} \right)^2 = 1 && (\text{ellipses, foci at } \pm 1), \\ y = \text{const} : \quad & \left( \frac{u}{\cos y} \right)^2 - \left( \frac{v}{\sin y} \right)^2 = 1 && (\text{hyperbolas, foci at } \pm 1). \end{aligned}$$

On the axes,

$$y = 0 : w = \cosh x \in [1, \infty), \quad x = 0 : w = \cos y \in [-1, 1].$$

**Critical points and injectivity on strips.**

$\cosh'(z) = \sinh z$ , which vanishes at  $z = i\pi k$ . Thus  $\cosh$  fails to be injective across lines  $y = k\pi$ , but is injective on any strip avoiding these lines (e.g.  $0 < \Im z < \pi$ ).

**Canonical strip mapping.**

$$\cosh : \{ 0 < \Im z < \pi \} \xrightarrow{\text{biholomorphism}} \mathbb{C} \setminus ((-\infty, -1] \cup [1, \infty)).$$

*Explanation.* The midline  $y = 0$  maps to  $[1, \infty)$ ; the top boundary  $y = \pi$  maps to  $(-\infty, -1]$ ; injectivity holds on  $0 < \Im z < \pi$  since there  $\sinh z \neq 0$ .

**Inverse branch and branch cuts.**

$$\operatorname{arcosh} w = \log \left( w + \sqrt{w-1} \sqrt{w+1} \right),$$

with branch cuts along  $(-\infty, -1] \cup [1, \infty)$  so that  $\Re(\operatorname{arcosh} w) \geq 0$  and  $0 < \Im(\operatorname{arcosh} w) < \pi$  on the principal sheet.

**Useful identities.**

$$\cosh(z + \zeta) = \cosh z \cosh \zeta + \sinh z \sinh \zeta, \quad \cosh z = 1 + 2 \sinh^2 \frac{z}{2}.$$

**3.  $w = \tanh z$** **Analytic background.**

$\tanh$  is *meromorphic*, *odd*, and  $i\pi$ -periodic:

$$\tanh(-z) = -\tanh z, \quad \tanh(z + i\pi) = \tanh z.$$

Zeros at  $z = i\pi k$ ; poles (simple) at  $z = i\pi(k + \frac{1}{2})$ .

**Strip  $\leftrightarrow$  disk (Cayley-type).**

$$\boxed{\tanh : \{ |\Im z| < \frac{\pi}{2} \} \xrightarrow{\text{biholomorphism}} \mathbb{D} .}$$

*Boundary correspondence.* The real axis  $y = 0$  maps bijectively onto  $(-1, 1)$  (strictly increasing). As  $y \rightarrow \pm \frac{\pi}{2}$ ,  $|\tanh(x + iy)| \rightarrow 1$  uniformly in  $x$ ; hence the horizontal boundaries map to  $\partial\mathbb{D}$ .

**Inverse (disk  $\rightarrow$  strip).**

$$\operatorname{artanh} w = \frac{1}{2} \log \left( \frac{1+w}{1-w} \right), \quad |w| < 1,$$

which maps  $\mathbb{D}$  conformally onto  $\{ |\Im z| < \frac{\pi}{2} \}$ .

**Derivative and Schwarz–Pick.**

$$(\tanh z)' = \operatorname{sech}^2 z, \quad |\tanh'(z)| = \frac{1}{|\cosh z|^2}$$

and conjugating by a disk automorphism shows  $\tanh$  is extremal for the disk-strip version of Schwarz–Pick.

**Level sets.**

Lines  $x = \text{const}$  and  $y = \text{const}$  in the strip map to orthogonal circle arcs inside  $\mathbb{D}$  meeting  $\partial\mathbb{D}$  orthogonally (images of preimages under the conformal equivalence with the half-plane followed by a Cayley map).

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**4.  $w = \coth z$** **Analytic background.**

$\coth$  is *meromorphic*, *odd*, and  $i\pi$ -periodic:

$$\coth(-z) = -\coth z, \quad \coth(z + i\pi) = \coth z.$$

Poles (simple) at  $z = i\pi k$ ; zeros (simple) at  $z = i\pi(k + \frac{1}{2})$ .

**Strip  $\leftrightarrow$  exterior disk.**

$$\boxed{\coth : \{ |\Im z| < \frac{\pi}{2} \} \setminus \{0\} \xrightarrow{\text{biholomorphism}} \{ |w| > 1 \} .}$$

*Boundary correspondence.* The real axis (excluding 0) maps to the rays  $(-\infty, -1) \cup (1, \infty)$ . As  $y \rightarrow \pm \frac{\pi}{2}$ ,  $|\coth(x + iy)| \rightarrow 1^+$ ; thus horizontal boundaries map to the unit circle from the *exterior* side.

**Inverse (exterior disk  $\rightarrow$  strip).**

$$\operatorname{arccoth} w = \frac{1}{2} \log \left( \frac{w+1}{w-1} \right), \quad |w| > 1,$$

with the standard logarithm branch so that  $\Im(\operatorname{arccoth} w) \in (-\frac{\pi}{2}, \frac{\pi}{2})$ .

**Relation with  $\tanh$ .**

$$\coth z = \tanh \left( z + \frac{i\pi}{2} \right),$$

so the exterior-disk picture is just the disk picture shifted vertically by  $i\pi/2$  and inverted via  $w \mapsto 1/w$ .

## 5. Global relations, inverse formulas, and composition patterns

**Inverse formulas (principal branches).**

$$\begin{aligned} \operatorname{arsinh} w &= \log(w + \sqrt{w^2 + 1}), & \operatorname{arcosh} w &= \log(w + \sqrt{w-1} \sqrt{w+1}), \\ \operatorname{artanh} w &= \frac{1}{2} \log \left( \frac{1+w}{1-w} \right), & \operatorname{arccoth} w &= \frac{1}{2} \log \left( \frac{w+1}{w-1} \right). \end{aligned}$$

Branch cuts are chosen to match the canonical target domains described above.

**Disk/half-plane equivalences via Cayley.**

Using the Cayley map  $C(\zeta) = \frac{1+\zeta}{1-\zeta}$  and its inverse  $C^{-1}(w) = \frac{w-1}{w+1}$ ,

$$\tanh z = C^{-1}(e^{2z}), \quad \coth z = C^{-1}(e^{2z})^{-1}, \quad \sinh z = \frac{e^z - e^{-z}}{2}, \quad \cosh z = \frac{e^z + e^{-z}}{2}.$$

Thus strip  $\leftrightarrow$  disk/exterior-disk mappings for  $\tanh$  and  $\coth$  are just the exponential's strip-to-sector equivalences composed with a Cayley map.

**Growth and boundary behavior.**

As  $x \rightarrow \pm\infty$  with  $y$  fixed,  $\sinh z \sim \frac{1}{2}e^x e^{iy}$  and  $\cosh z \sim \frac{1}{2}e^x e^{iy}$ ; hence horizontal lines map to asymptotically radial curves. For  $\tanh$  and  $\coth$ ,  $|\tanh(x+iy)| \rightarrow 1$  and  $|\coth(x+iy)| \rightarrow 1$  exponentially in  $|x|$  (uniform in  $y$  within the strip height).

**Critical sets and injectivity regions.**

$\sinh'(z) = \cosh z$  vanishes along  $y = \frac{\pi}{2} + \pi\mathbb{Z}$ ;  $\cosh'(z) = \sinh z$  vanishes along  $y = \pi\mathbb{Z}$ . Therefore, the natural injectivity strips are  $|\Im z| < \frac{\pi}{2}$  for  $\sinh$  and  $0 < \Im z < \pi$  for  $\cosh$ . For  $\tanh$  and  $\coth$ , poles on the horizontal boundaries determine the natural strip height  $\pi$ .

## 6. Summary: canonical conformal equivalences (boxed)

$$\sinh : \{ |\Im z| < \frac{\pi}{2} \} \longrightarrow \mathbb{C} \setminus i((-\infty, -1] \cup [1, \infty)) \quad (\text{biholomorphism}) .$$

$$\cosh : \{ 0 < \Im z < \pi \} \longrightarrow \mathbb{C} \setminus ((-\infty, -1] \cup [1, \infty)) \quad (\text{biholomorphism}) .$$

$$\tanh : \{ |\Im z| < \frac{\pi}{2} \} \longrightarrow \mathbb{D} \quad (\text{biholomorphism}) .$$

$$\coth : \{ |\Im z| < \frac{\pi}{2} \} \setminus \{0\} \longrightarrow \{ |w| > 1 \} \quad (\text{biholomorphism}) .$$

## 7. (Optional) Proof sketches for strip mappings

**Injectivity for  $\sinh$  on  $|\Im z| < \pi/2$ .** If  $\sinh z_1 = \sinh z_2$  then  $e^{z_1} - e^{-z_1} = e^{z_2} - e^{-z_2}$ . Rearranging gives  $\{e^{z_1}, e^{-z_1}\} = \{e^{z_2}, e^{-z_2}\}$ . Taking logs in the horizontal strip  $|\Im z| < \pi/2$  (where the exponential is injective modulo  $2\pi i$ ) forces  $z_1 = z_2$ .

**Boundary images for  $\sinh$ .** Set  $z = x \pm i\pi/2$ . Then  $\sinh z = \sinh x \cdot 0 \pm i \cosh x$ , so the boundaries map to  $\pm i[1, \infty)$ ; the interior cannot reach those rays by the open mapping theorem and the maximum modulus principle applied to  $1/(\sinh z \mp i)$ .

**Analogous arguments** apply to  $\cosh$  (use  $\sinh$  zeros to avoid critical lines) and to  $\tanh, \coth$  using the representation via exponentials and the Cayley transform.

## Inverse Circular Functions: arcsin, arccos, arctan, arccot

### Conventions

We use principal branches of  $\log$  and  $\sqrt{\cdot}$  with branch cut  $(-\infty, 0]$  unless stated otherwise. For each inverse, the principal domain is obtained by slitting the  $w$ -plane along the images of the multiple values.

1.  $z = \arcsin w$

**Principal formula (single-valued on the slit plane).**

$$\arcsin w = -i \log \left( i w + \sqrt{1 - w^2} \right)$$

**Branch points & cuts.**

Branch points at  $w = \pm 1$ ; principal cuts along  $(-\infty, -1] \cup [1, \infty)$ .

**Target strip (canonical range).**

$$\arcsin : \mathbb{C} \setminus ((-\infty, -1] \cup [1, \infty)) \longrightarrow \{ |\Re z| < \frac{\pi}{2} \}$$

For  $w \in (-1, 1)$ ,  $\arcsin w \in (-\frac{\pi}{2}, \frac{\pi}{2}) \subset \mathbb{R}$ . For  $w > 1$ ,  $\arcsin w = \frac{\pi}{2} - i \operatorname{arcosh} w$  (hence  $\Re = \frac{\pi}{2}$ ); for  $w < -1$ ,  $\arcsin w = -\frac{\pi}{2} + i \operatorname{arcosh}(-w)$  (hence  $\Re = -\frac{\pi}{2}$ ).

**Boundary correspondence.**

Approaching the cuts  $[1, \infty)$  or  $(-\infty, -1]$  forces  $\Re \arcsin w \rightarrow \pm \frac{\pi}{2}$  and  $|\Im \arcsin w| \rightarrow \infty$ . The real axis segment  $(-1, 1)$  maps to the real segment  $(-\frac{\pi}{2}, \frac{\pi}{2})$ .

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**2.  $z = \arccos w$** **Principal formulae.**

$$\arccos w = \frac{\pi}{2} - \arcsin w$$

$$\arccos w = \frac{1}{i} \log(w + \sqrt{w-1} \sqrt{w+1})$$

**Branch points & cuts.**

Same as arcsin: branch points  $\pm 1$ ; cuts  $(-\infty, -1] \cup [1, \infty)$ .

**Target strip (canonical range).**

$$\arccos : \mathbb{C} \setminus ((-\infty, -1] \cup [1, \infty)) \longrightarrow \{ 0 < \Re z < \pi \}$$

For  $w \in (-1, 1)$ ,  $\arccos w \in (0, \pi) \subset \mathbb{R}$ . For  $w > 1$ ,  $\arccos w = i \operatorname{arcosh} w$  (so  $\Re = 0$ ); for  $w < -1$ ,  $\arccos w = \pi - i \operatorname{arcosh}(-w)$  (so  $\Re = \pi$ ).

**Boundary correspondence.**

The two slits map to the vertical strip boundaries  $\Re z = 0$  and  $\Re z = \pi$ . The central interval  $(-1, 1)$  maps to the real segment  $(0, \pi)$ .

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**3.  $z = \arctan w$** **Principal formula.**

$$\arctan w = \frac{1}{2i} \log\left(\frac{1+iw}{1-iw}\right)$$

**Branch points & cuts.**

Branch points at  $w = \pm i$ ; principal cuts  $i(-\infty, -1] \cup i[1, \infty)$  along the imaginary axis beyond  $\pm i$ .

**Target strip (canonical range).**

$$\arctan : \mathbb{C} \setminus i((-\infty, -1] \cup [1, \infty)) \longrightarrow \{|\Re z| < \frac{\pi}{2}\}$$

On  $\mathbb{R}$ ,  $\arctan \mathbb{R} = (-\frac{\pi}{2}, \frac{\pi}{2})$ . Approaching  $w = \pm i t$  ( $t \geq 1$ ) drives  $\Re \arctan w \rightarrow \pm \frac{\pi}{2}$  and  $|\Im| \rightarrow \infty$ .

**Symmetries.**

$\arctan(-w) = -\arctan w$ ; conjugation across the real axis preserves values away from cuts.

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**4.  $z = \operatorname{arccot} w$** **Principal formulae.**

$$\operatorname{arccot} w = \frac{1}{2i} \log \left( \frac{w+i}{w-i} \right)$$

$$\operatorname{arccot} w = \frac{\pi}{2} - \arctan w$$

**Branch points & cuts.**

As for  $\arctan$ : branch points at  $w = \pm i$ ; cuts  $i(-\infty, -1] \cup i[1, \infty)$ .

**Target strip (canonical range).**

$$\operatorname{arccot} : \mathbb{C} \setminus i((-\infty, -1] \cup [1, \infty)) \longrightarrow \{0 < \Re z < \pi\}$$

On  $\mathbb{R}$ ,  $\operatorname{arccot} \mathbb{R} = (0, \pi)$ . Approaching the cuts pins the image to  $\Re z = 0$  or  $\Re z = \pi$  with  $|\Im| \rightarrow \infty$ .

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**5. Relations and identities (principal branches)**

$$\arcsin w + \arccos w = \frac{\pi}{2}, \quad \arctan w + \operatorname{arccot} w = \frac{\pi}{2}.$$

$$\arcsin w = -i \operatorname{arsinh}(iw), \quad \arccos w = \operatorname{arcosh} w \cdot \frac{1}{i} \quad (\text{on } [1, \infty)).$$

$$\arctan w = \frac{1}{2i} (\log(1+iw) - \log(1-iw)), \quad \operatorname{arccot} w = \frac{1}{2i} (\log(w+i) - \log(w-i)).$$

## 6. Boundary/axis behavior (summary)

- arcsin: real on  $(-1, 1)$ ; the cuts map to  $\Re z = \pm \frac{\pi}{2}$ .
- arccos: real on  $(-1, 1)$ ; the cuts map to  $\Re z = 0, \pi$ .
- arctan: real on  $\mathbb{R}$ ; the imaginary-axis cuts beyond  $\pm i$  map to  $\Re z = \pm \frac{\pi}{2}$ .
- arccot: real on  $\mathbb{R}$ ; the same cuts map to  $\Re z = 0, \pi$ .

## Conformal maps: an overview catalogue (no details)

### 1. Cayley (disk $\leftrightarrow$ right half-plane).

$$\boxed{w = \frac{1+z}{1-z}} \quad \text{inverse: } z = \frac{w-1}{w+1}.$$

### 2. Half-plane $\rightarrow$ half-plane (three-point normalization).

$$\boxed{w = \frac{(z-a)(c-b)}{(z-b)(c-a)}}.$$

### 3. Disk automorphisms.

$$\boxed{\phi_{a,\theta}(z) = e^{i\theta} \frac{z-a}{1-\bar{a}z}}, \quad |a| < 1.$$

### 4. Strip $|\Im z| < h/2 \rightarrow$ right half-plane.

$$\boxed{w = \exp\left(\frac{\pi z}{h}\right)}.$$

### 5. Strip $|\Im z| < h/2 \rightarrow$ unit disk.

$$\boxed{\zeta = \frac{e^{\pi z/h} - 1}{e^{\pi z/h} + 1}}.$$

### 6. Strip $\rightarrow$ strip (height scaling).

$$\boxed{z \mapsto \frac{h_2}{h_1} z}.$$

### 7. Band $a < \Im z < b \rightarrow$ centered strip.

$$\boxed{z \mapsto \frac{\pi}{b-a} \left( z - i \frac{a+b}{2} \right)}.$$

### 8. Sector $\{0 < \arg z < \alpha\} \rightarrow$ upper half-plane.

$$\boxed{w = z^{\pi/\alpha}} \quad (\text{principal branch}).$$

9. **Wedge**  $\rightarrow$  **disk**.

Compose item (8) with Cayley (item 1).

10. **Upper half-plane**  $\rightarrow$  **disk (prescribed triple)**.

$$\Phi(z) = \frac{(z-a)}{(z-\bar{a})} \cdot \frac{(b-\bar{a})}{(b-a)}.$$

11. **Annulus**  $r < |z| < 1 \leftrightarrow$  **strip**.

$$\boxed{w = \log z} \quad \text{maps to} \quad \log r < \Re w < 0.$$

12. **Annulus**  $\rightarrow$  **disk**.

Use  $w = \log z$  then item (5).

13. **Half-plane minus disk**  $\rightarrow$  **annulus**.

$$\boxed{\zeta = \frac{z-\alpha}{z+\alpha}}, \quad \alpha^2 = a^2 - b^2.$$

14. **Disk with off-center hole**  $\rightarrow$  **annulus**.

Center via  $\phi_a$  (item 3), then items (11)–(12).

15. **Punctured plane**  $\leftrightarrow$  **cylinder**.

$$\boxed{w = \log z} \quad (\text{period } 2\pi i).$$

16. **Strip**  $\rightarrow$  **punctured plane**.

$$\boxed{z \mapsto e^z}.$$

17. **Ray-slit plane**  $\mathbb{C} \setminus [0, \infty) \rightarrow$  **half-plane**.

$$\boxed{w = \sqrt{z}} \quad (\text{choose branch}).$$

18. **Segment-slit plane**  $\mathbb{C} \setminus [a, b] \rightarrow$  **half-plane**.

$$\boxed{w = \sqrt{\frac{z-a}{z-b}}}.$$

19. **Exterior of**  $[-1, 1] \leftrightarrow$  **exterior unit circle (Joukowski)**.

$$\boxed{w = z + \frac{1}{z}}, \quad z = \frac{w \pm \sqrt{w^2 - 4}}{2}.$$

20. **Exterior of ellipse**  $\leftrightarrow$  **exterior unit circle**.

$$\boxed{w = \frac{1}{2} \left( cz + \frac{1}{cz} \right)} \quad (c \text{ from axes}).$$

21. **Right half-plane**  $\rightarrow$  **vertical strip**.

$$\boxed{z \mapsto \frac{1}{\pi} \log \frac{z-1}{z+1}}.$$

22. **Upper half-plane  $\rightarrow$  rectangle (SC/elliptic).**

$$\zeta = \int^z \frac{dt}{\sqrt{(t-a)(t-b)(t-c)(t-d)}}.$$

23. **Rectangle  $\rightarrow$  disk (Jacobi sn).**

$$\zeta = \operatorname{sn}\left(\frac{K(k)}{L} z; k\right).$$

24. **Half-disk  $\leftrightarrow$  half-plane.**

Restriction of Cayley (item 1) and inverse.

25. **Lens (intersection of two disks)  $\rightarrow$  disk.**

Möbius normalizes the circles; then a power map; then Cayley.

26. **Half-strip  $\{x > 0, 0 < \Im z < h\} \rightarrow$  half-disk.**

$$\exp(\pi z/h) \quad \text{then power, then Cayley.}$$

27. **Polygon  $\rightarrow$  half-plane (Schwarz–Christoffel).**

$$\Phi'(z) = C \prod_k (z - z_k)^{\alpha_k - 1}, \quad \sum \alpha_k = 2.$$

28. **Upper half-plane  $\rightarrow$  slit half-plane.**

$$w = z + \frac{1}{z} \quad \text{then a real Möbius adjustment.}$$

29. **Disk  $\rightarrow$  disk with boundary arc removed.**

Automorphism to put endpoints at  $\pm 1$ , then boundary power  $z^\lambda$ .

30. **Finite Blaschke product (disk  $\rightarrow$  disk, degree  $n$ ).**

$$B(z) = e^{i\theta} \prod_{k=1}^n \frac{z - a_k}{1 - \bar{a}_k z}.$$

## 1. Cayley transform: $\mathbb{D} \leftrightarrow$ right half-plane

Map and inverse.

$$w = \frac{1+z}{1-z} \quad \text{maps} \quad \mathbb{D} = \{|z| < 1\} \quad \text{onto} \quad \{\Re w > 0\}, \quad z = \frac{w-1}{w+1}$$

is the inverse.

**Why  $\Re w > 0$ .**

Compute

$$\Re\left(\frac{1+z}{1-z}\right) = \frac{1-|z|^2}{|1-z|^2}.$$

Thus  $\Re w > 0$  for  $|z| < 1$ , and  $\Re w = 0$  for  $|z| = 1$  (except  $z = 1$ , which maps to  $w = \infty$ ).  
Therefore

$$|z| < 1 \iff \Re w > 0, \quad |z| = 1 \iff \Re w = 0.$$

**Boundary correspondence.**

$z = 0 \mapsto w = 1$ ,  $z \rightarrow 1^- \mapsto w \rightarrow +\infty$ , and  $\partial\mathbb{D} \setminus \{1\}$  maps to the line  $\Re w = 0$ .

Circles/lines orthogonal to  $\partial\mathbb{D}$  map to *vertical* lines in the  $w$ -plane.

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## 2. Three-point Möbius normalization: half-plane $\leftrightarrow$ half-plane

**Cross-ratio map to the real axis.**

Let  $L$  be a line or circle, and choose three distinct points  $a, b, c \in L$ . Define

$$\Phi(z) = \frac{(z-a)(c-b)}{(z-b)(c-a)}.$$

Then  $\Phi(L) \subset \mathbb{R} \cup \{\infty\}$ , with

$$\Phi(a) = 0, \quad \Phi(b) = \infty, \quad \Phi(c) = 1,$$

and the two sides of  $L$  map to the half-planes  $\Im \Phi(z) \gtrless 0$ .

**Half-plane to half-plane via three boundary points.**

Given half-planes  $H_1, H_2$  with boundary arcs  $L_1, L_2$ , pick ordered boundary triples  $a, b, c \in L_1$  and  $A, B, C \in L_2$  (same cyclic order). Then

$$T(z) = \frac{(z-a)(c-b)}{(z-b)(c-a)} \cdot \frac{(C-A)}{(C-B)}$$

is a Möbius map sending  $L_1 \rightarrow L_2$  with

$$a \mapsto A, \quad b \mapsto B, \quad c \mapsto C,$$

and taking the chosen side of  $H_1$  onto the chosen side of  $H_2$ .

**Specialization to the real line.**

To send  $L$  to  $\mathbb{R}$ , use  $\Phi$  above; then pre/postcompose with an *affine real* map

$$\xi \mapsto \alpha \xi + \beta, \quad \alpha > 0,$$

to obtain any target half-plane, e.g.  $\Re w > 0$  or  $\Im w > 0$ .